

DIGITAL RADIO LAB

Implementation of Digital Modulation & Demodulation Techniques (OOK, ASK, PSK, QPSK & 16-Level Modulation) using GNU Radio and ADALM-PLUTO SDR

LABORATORY REPORT

Detail	Information
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1. Aim

To implement digital modulation and demodulation schemes — On-Off Keying (OOK), Amplitude Shift Keying (ASK), Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK) and a 16-level (higher order) modulation scheme — using GNU Radio Companion (GRC), and to demonstrate real-time RF transmission and reception between two ADALM-PLUTO Software Defined Radio (SDR) units. The received signals are to be analysed in the time domain and frequency domain, and the different modulation schemes are to be compared on the basis of their constellation diagrams.

2. Software & Hardware Used

Item	Description
Software	GNU Radio Companion (GRC) — graphical, block-based (drag-and-drop) SDR development environment
SDR Hardware	Two ADALM-PLUTO (PlutoSDR) units — one configured as Transmitter, one as Receiver
Interface	PlutoSDR Source / PlutoSDR Sink blocks (IIO context over Ethernet/USB, e.g. ip:192.168.2.1, ip:192.168.99.1)
GUI Sinks	QT GUI Time Sink, QT GUI Frequency Sink, QT GUI Constellation Sink, QT GUI Range, QT GUI Chooser
Operating System	Windows / Linux host running GNU Radio

Note: The entire signal chain (transmitter and receiver) was built purely using GRC's drag-and-drop block canvas. No block was hand-coded in Python; GRC auto-generates the underlying flow-graph code when the design is executed.

3. Theory

3.1 Digital Modulation

Digital modulation is the process of mapping a stream of digital bits onto variations of a carrier wave's amplitude, phase, or frequency so that the information can be transmitted over an RF channel. At the receiver, the reverse process (demodulation) recovers the original bit stream from the received carrier.

3.2 On-Off Keying (OOK)

OOK is the simplest form of ASK in which a binary '1' is represented by the presence of the carrier (full amplitude) and a binary '0' by its absence (zero amplitude). It is easy to generate and detect (using an envelope/threshold detector) but is highly sensitive to noise and amplitude fading.

3.3 Amplitude Shift Keying (ASK)

In ASK, the amplitude of the carrier is varied between two (or more) discrete, generally non-zero, levels according to the data bits, while frequency and phase are kept constant. Demodulation is performed using envelope detection or coherent detection followed by a decision (threshold) device.

3.4 Phase Shift Keying — BPSK and QPSK

In Binary Phase Shift Keying (BPSK), the carrier phase is switched between two values 180° apart (e.g., 0° and 180°) to represent bit '0' and bit '1'. BPSK gives the best noise immunity among binary schemes because the two symbols are maximally separated in phase. Quadrature Phase Shift Keying (QPSK) extends this idea by

using four phase states (45° , 135° , 225° , 315°), each representing two bits (a dibit). This doubles the bit rate for the same symbol rate/bandwidth compared to BPSK, at the cost of reduced noise margin between adjacent symbols.

3.5 Higher-Order Modulation (16-Level Scheme)

To increase spectral efficiency further, multiple bits can be mapped onto a single symbol using more amplitude/phase levels — for example 16-PSK, 16-QAM or 16-PAM place 16 distinct symbols in the signal space, each carrying 4 bits per symbol. In this experiment a 16-point symbol constellation (mapped through a Chunks-to-Symbols block with a 16-point complex symbol table) was implemented in GRC as a higher-order scheme analogous to 16-QAM/16-PAM, to study the trade-off between spectral efficiency and noise/impairment sensitivity as the number of constellation points increases.

3.6 Constellation Diagram

A constellation diagram plots the in-phase (I) component of a symbol on the x-axis and the quadrature (Q) component on the y-axis. Each modulation scheme produces a characteristic pattern of points: OOK/ASK produce points along the real axis (varying amplitude), BPSK shows 2 points at 180° separation, QPSK shows 4 points at 90° separation on a circle, and higher-order schemes (16-level) show 16 points arranged in a grid or circle. The tightness/spread of the points around the ideal locations indicates the noise, phase-offset and I/Q imbalance present in the received signal.

3.7 Time Domain and Frequency Domain Analysis

The QT GUI Time Sink displays the instantaneous amplitude of the I and Q signal components versus time, useful for observing the modulation envelope/phase transitions. The QT GUI Frequency Sink computes and displays the FFT of the signal (Power Spectral Density vs. frequency), which shows the occupied bandwidth, the carrier peak and any spectral regrowth or interference.

4. Source Code (GRC Flow-graph Blocks)

As this laboratory work was carried out entirely inside GNU Radio Companion using its drag-and-drop graphical blocks (no manual Python coding was performed), this section lists the key functional blocks used to build the Transmitter (Tx) and Receiver (Rx) flow-graphs for each scheme, instead of raw source code. The corresponding block diagrams are provided in Section 6 (Observations).

4.1 Common Blocks

Block	Function
Vector Source / Signal Source	Generates the baseband bit pattern or a reference tone (e.g., cosine wave) used as the data/carrier reference
Chunks to Symbols	Maps input bits/symbols to complex constellation points using a user-defined Symbol Table
Repeat / Interpolation	Up-samples each symbol to the desired samples-per-symbol (sps) so it can be transmitted through the PlutoSDR
Multiply	Mixes/modulates the symbol stream with a carrier / combines signal paths
PlutoSDR Sink	Transmits the generated baseband IQ samples over the air through the PlutoSDR TX chain (sets LO frequency, sample rate, buffer size, Tx attenuation, RF bandwidth)
PlutoSDR Source	Receives IQ samples from the PlutoSDR RX chain (sets LO frequency, sample rate, RX gain, RF bandwidth)
QT GUI Time Sink	Displays received/transmitted signal amplitude vs. time
QT GUI Frequency Sink	Displays the power spectrum (FFT) of the signal
QT GUI Constellation Sink	Plots the I-Q scatter (constellation) of the signal
Complex to Mag / Threshold / Keep 1 in N	Used in the OOK receiver chain for envelope detection and bit decision
AGC, Constellation Decoder, Char to Float	Used in the ASK/QAM receiver chains for automatic gain control and symbol decision

4.2 Scheme-wise Flow-graph Titles (as created in GRC)

GRC Project Title	Scheme Implemented	Role
Pluto_waveform	Baseline waveform generation & loop-back check (Sine/Cosine/Square/Triangle/Sawtooth)	Preliminary hardware verification
FM Reception	WBFM broadcast reception (reference/calibration test)	Preliminary hardware verification
OOK Group 1	On-Off Keying	Tx + Rx
PSK Group 1	Binary Phase Shift Keying (BPSK)	Tx + Rx
QPSK Group 1	Quadrature Phase Shift Keying	Tx + Rx
ASK Group 1	Amplitude Shift Keying	Tx + Rx
PSK_16 Group 1	16-level (higher order) modulation, 16-point complex symbol table	Tx + Rx

5. Procedure

- Connect two ADALM-PLUTO SDR units to the host PC(s) — one is configured as the Transmitter (Tx) node and the other as the Receiver (Rx) node.
- Verify each PlutoSDR's IIO context address (e.g., ip:192.168.2.1 for Tx, ip:192.168.99.1 for Rx) and confirm connectivity in GNU Radio Companion.
- Perform a preliminary hardware check by generating standard test waveforms (Sine, Cosine, Square, Triangle, Sawtooth) through the PlutoSDR Sink and observing them back through the PlutoSDR Source, and by receiving a live FM broadcast station, to confirm that both SDR units and the RF link are functioning correctly.
- For each digital modulation scheme (OOK, ASK, BPSK, QPSK, 16-level), build the Transmitter flow-graph: generate the bit/symbol sequence with a Vector Source, map it to constellation points using Chunks to Symbols with the appropriate Symbol Table, up-sample using the Repeat block, and send the samples to the PlutoSDR Sink at a common LO/centre frequency.
- Build the corresponding Receiver flow-graph on the second PlutoSDR: capture RF samples with the PlutoSDR Source at the same centre frequency, and connect QT GUI Time, Frequency and Constellation Sinks to visualise the received signal. For OOK, add envelope detection (Complex to Mag) and a Threshold block for bit recovery; for ASK/16-level schemes, add an AGC and Constellation Decoder for symbol recovery.
- Set the Tx attenuation and Rx gain using the QT GUI Range sliders to obtain a clean, detectable received signal.
- Execute the Transmitter and Receiver flow-graphs simultaneously (Tx first, then Rx) so that real-time RF transmission and reception takes place over the air between the two PlutoSDR units.
- Observe and record the transmitted and received signals in the time domain (QT GUI Time Sink) and frequency domain (QT GUI Frequency Sink) for every scheme.
- Observe and record the constellation diagram at the receiver for every scheme and compare the number/arrangement of constellation points, and the amount of scatter/spread around each ideal point, across the different modulation schemes.
- Repeat the above steps for OOK, ASK, BPSK, QPSK and the 16-level scheme, keeping the measurement procedure identical for a fair comparison.

6. Observations

The screenshots below were captured directly from the GNU Radio Companion flow-graphs and their live QT GUI output windows during real-time transmission and reception between the two PlutoSDR units. For every scheme, the Transmitter (Tx) side outputs are presented first, followed by the Receiver (Rx) side outputs.

6.1 On-Off Keying (OOK)

Transmitter Side

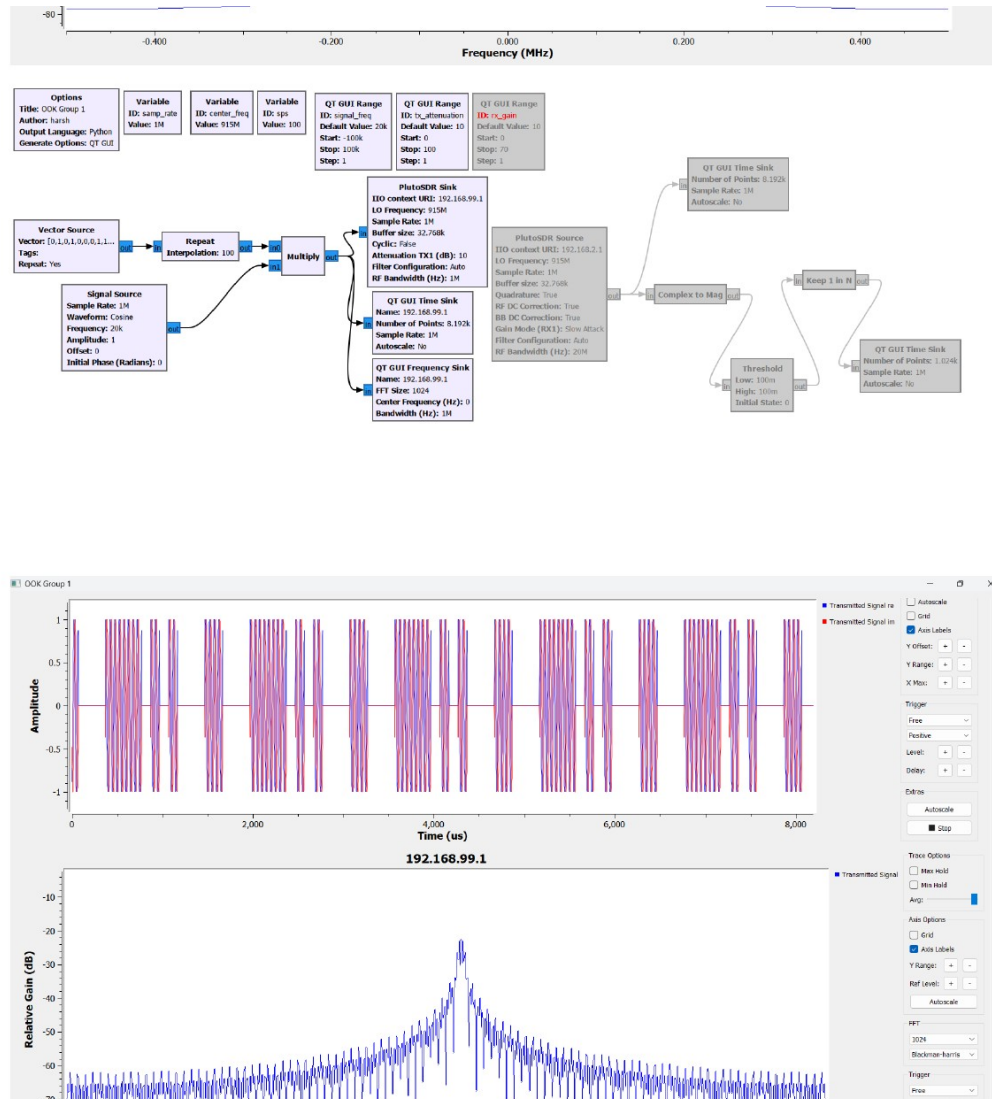


Fig 6.1(a): OOK Group 1 — GRC flow-graph and transmitted OOK signal in the time domain (top) with its power spectrum (bottom)

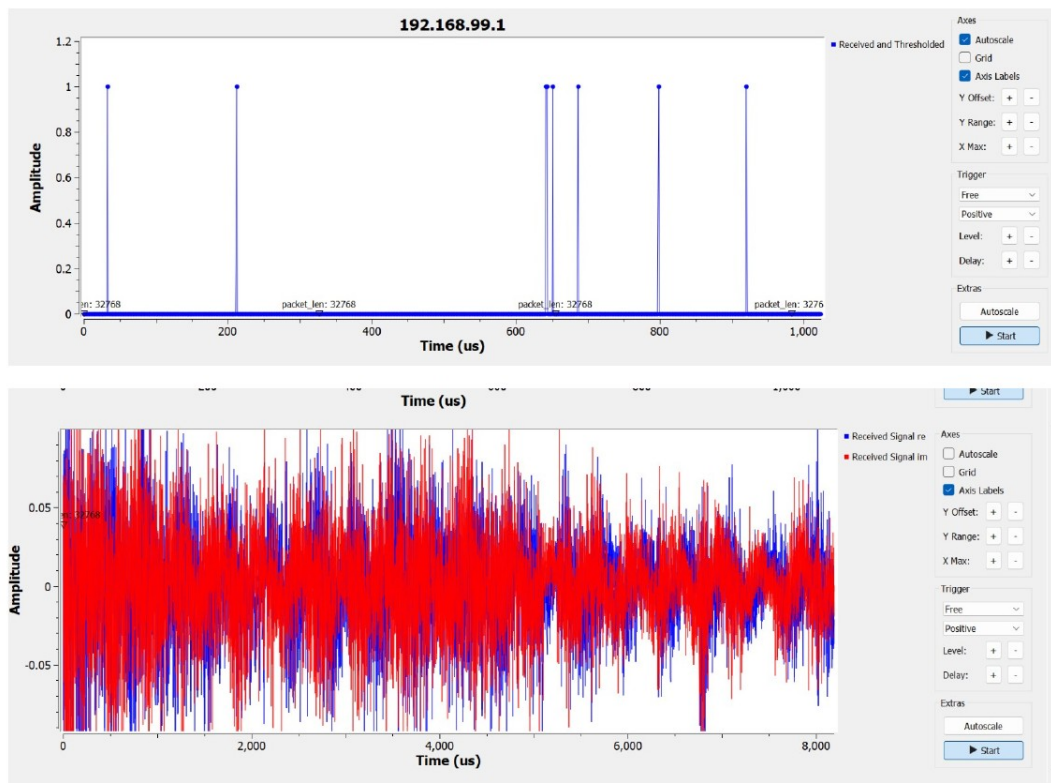
Receiver Side

Fig 6.1(b): OOK Receiver — thresholded/decoded output pulses (top) and received baseband I/Q waveform (bottom)

6.2 Binary Phase Shift Keying (BPSK)

Transmitter Side

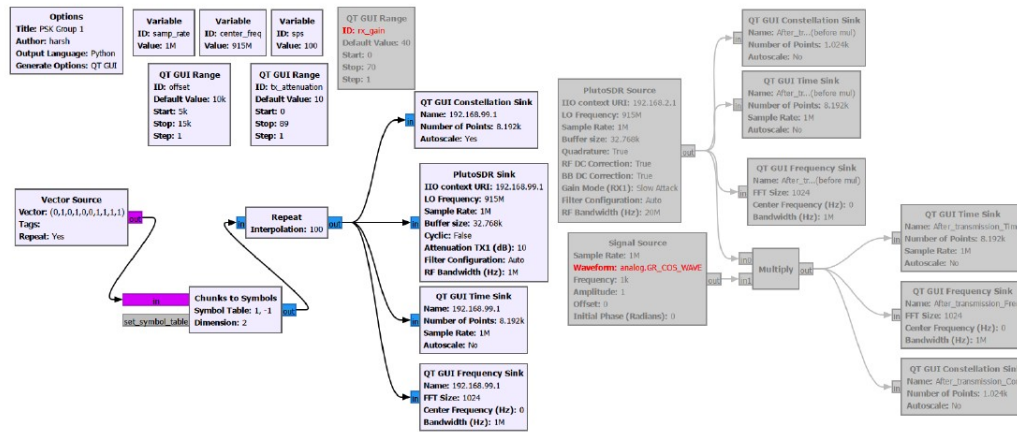


Fig 6.2(a): PSK Group 1 — GRC flow-graph for BPSK transmission and reception

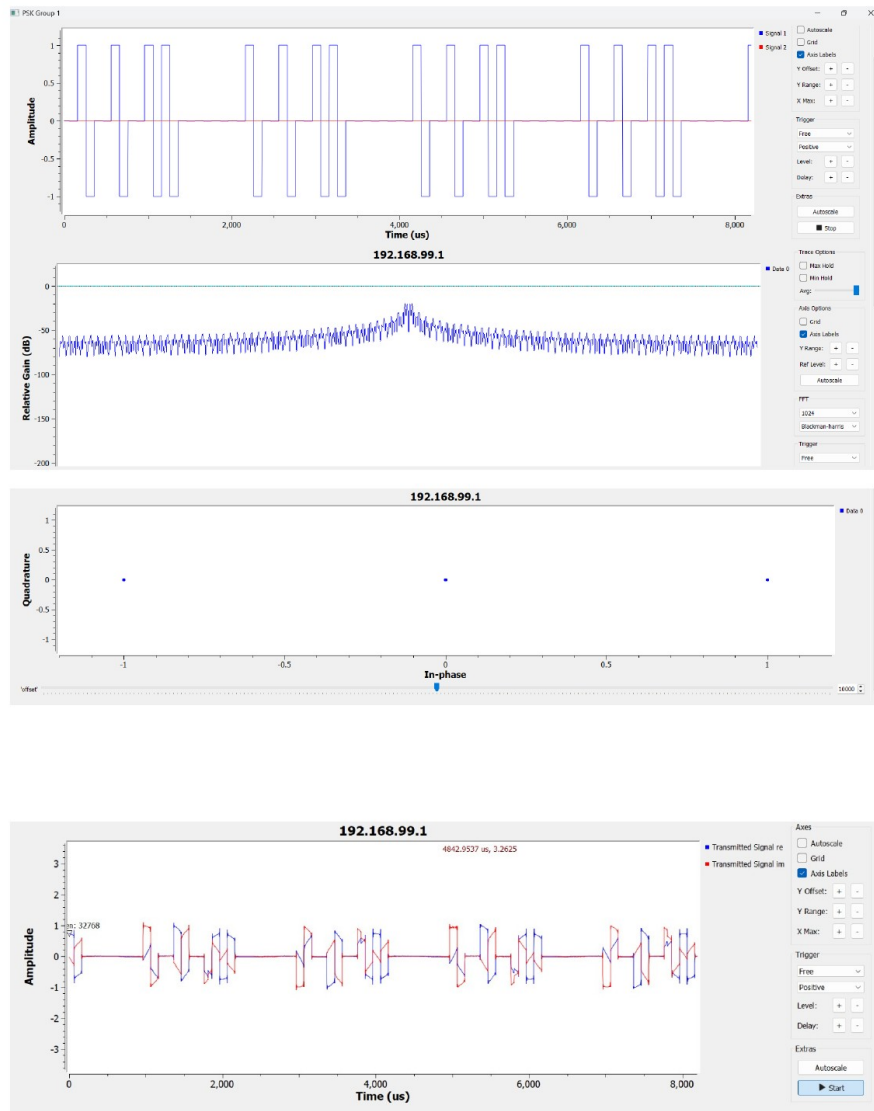


Fig 6.2(b): BPSK baseband signals (top), received power spectrum (middle), constellation before modulation (lower-middle) and transmitted I/Q waveform (bottom)

Receiver Side

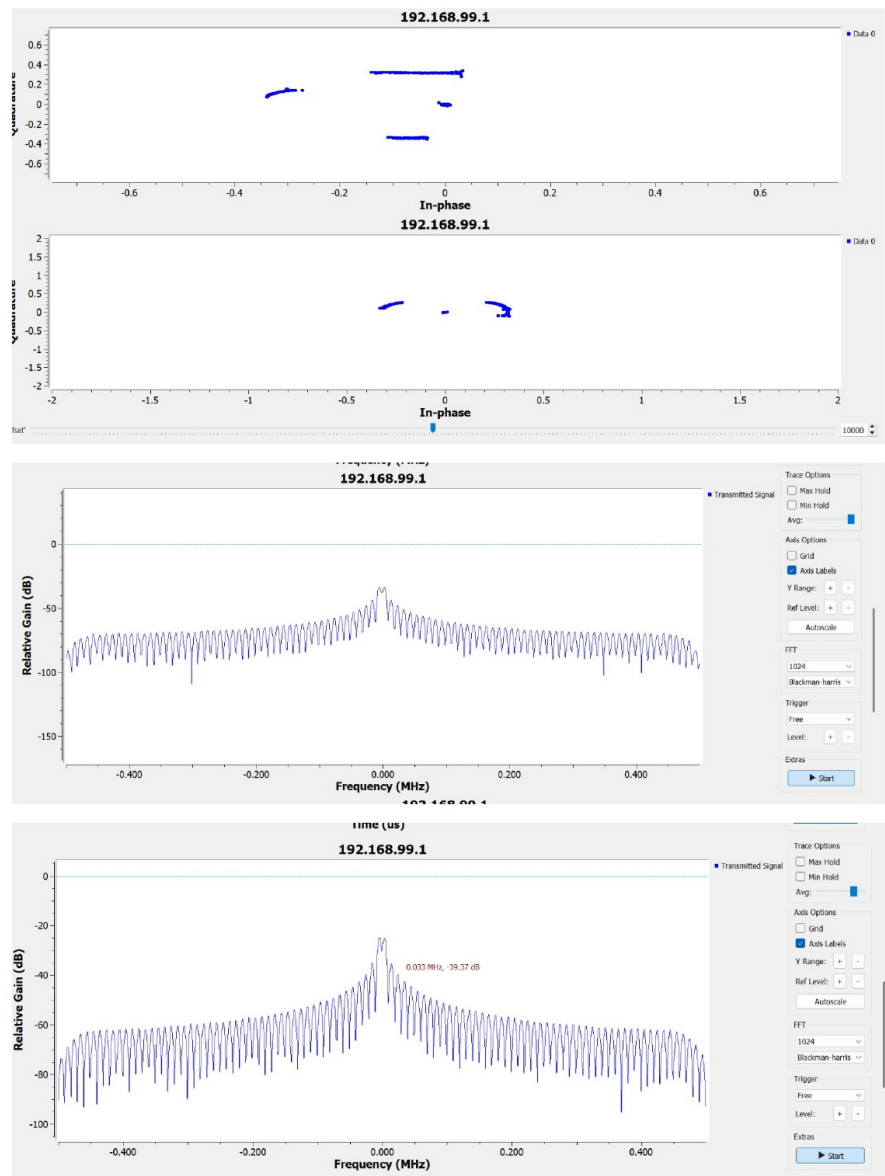


Fig 6.2(c): BPSK receiver — I/Q constellation scatter (top two) and received signal power spectrum (bottom two)

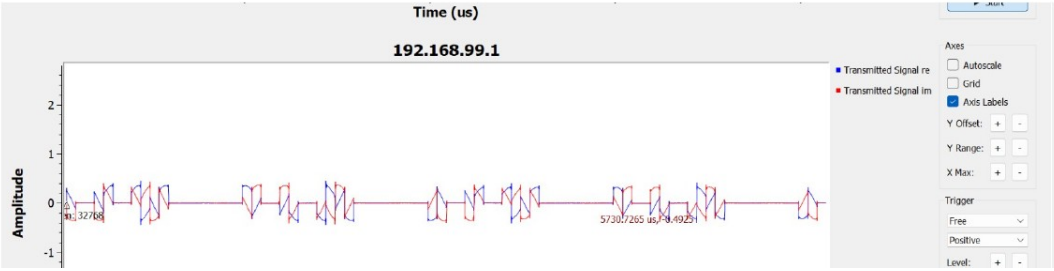


Fig 6.2(d): BPSK transmitted signal waveform observed after transmission (receiver side)

6.3 Quadrature Phase Shift Keying (QPSK)

Transmitter Side

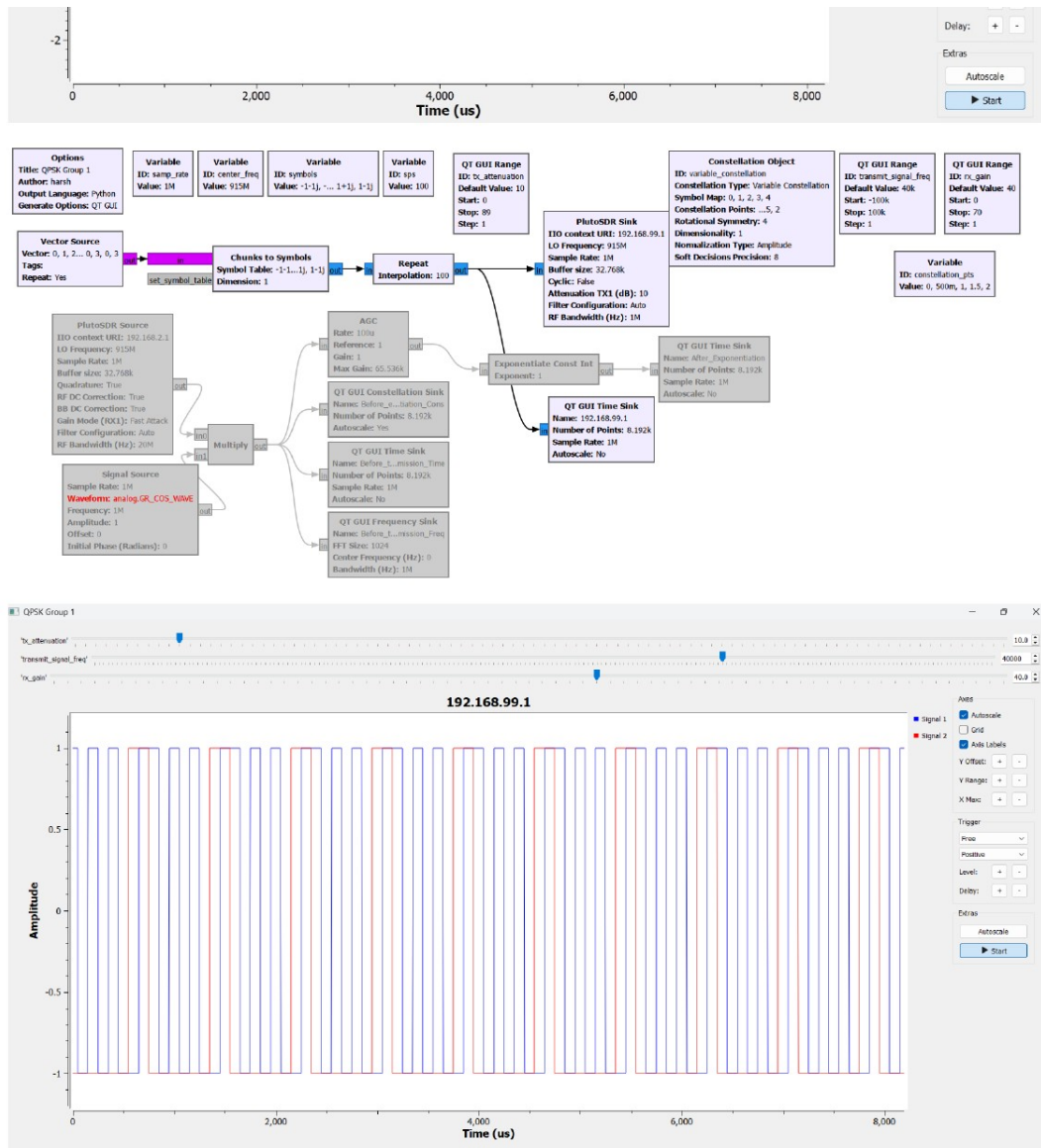


Fig 6.3(a): QPSK Group 1 — GRC flow-graph with Tx attenuation / frequency / Rx gain control sliders, and baseband Signal 1 & 2

Receiver Side

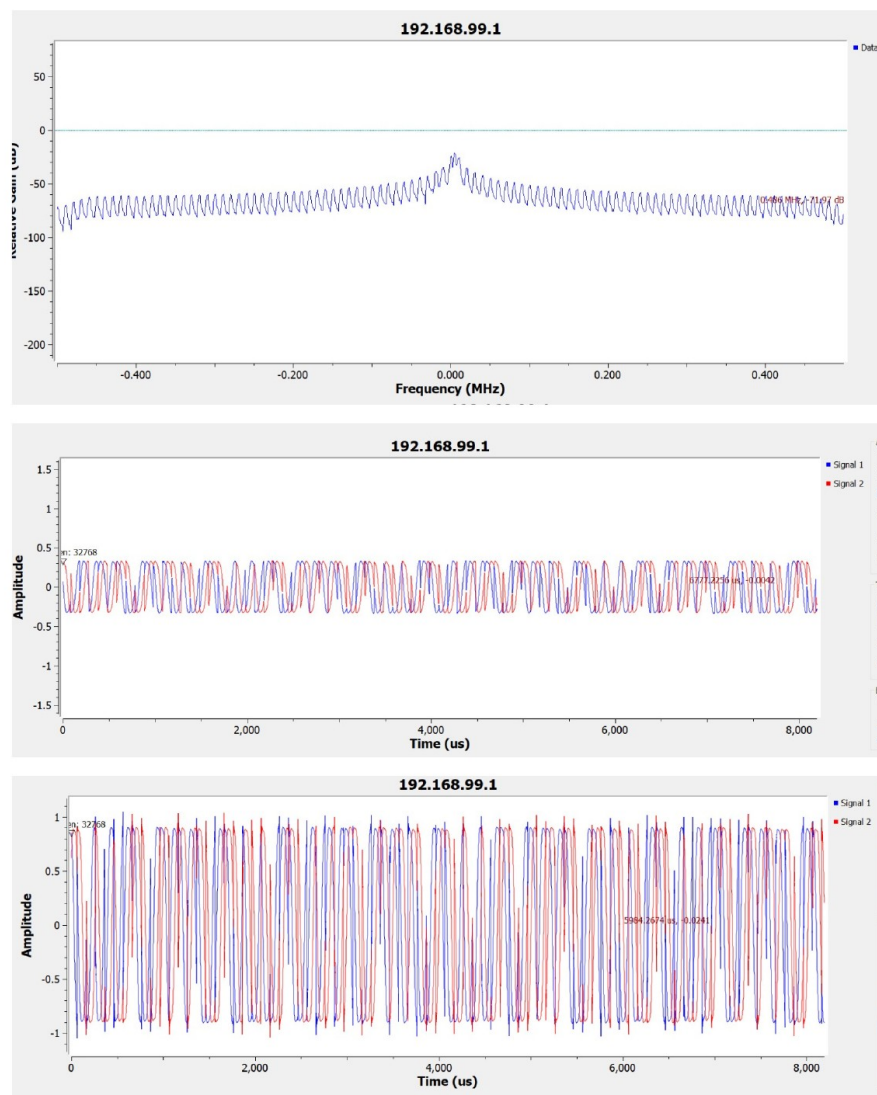


Fig 6.3(b): QPSK received power spectrum (top) and decision-stage Signal 1 & 2 waveforms (middle, bottom)

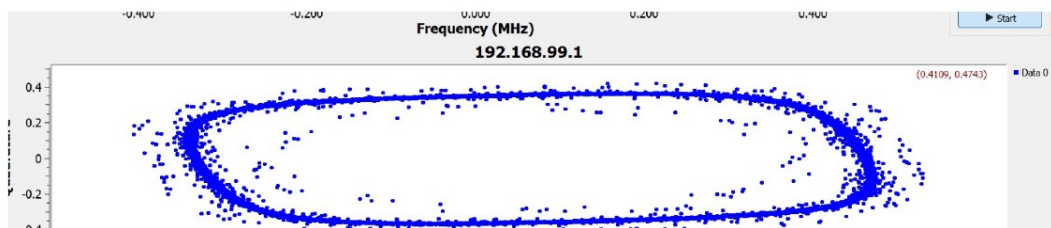


Fig 6.3(c): Final received QPSK constellation diagram

6.4 Amplitude Shift Keying (ASK)

Transmitter Side

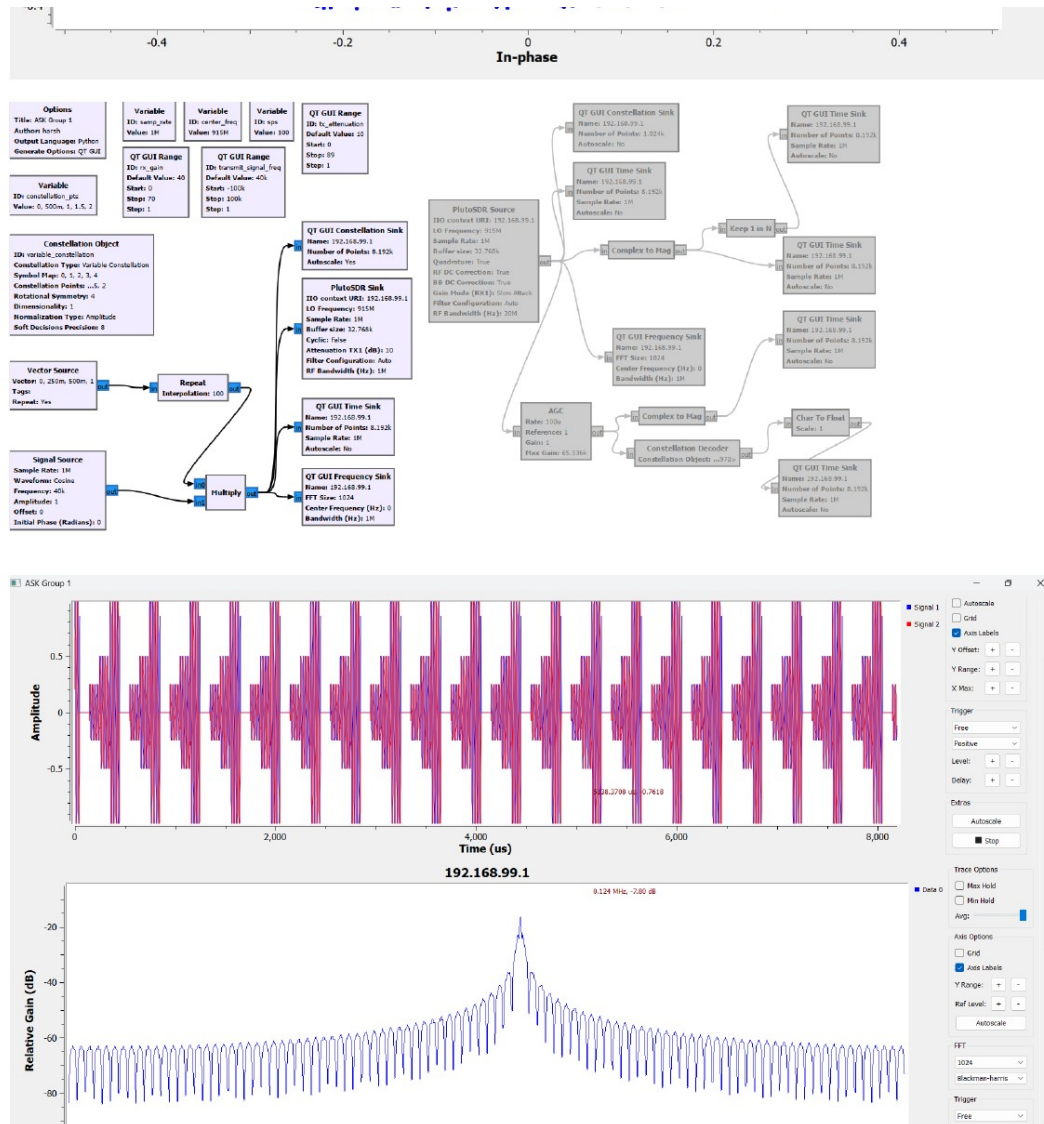


Fig 6.4(a): ASK Group 1 — GRC flow-graph and transmitted baseband Signal 1 & 2 with power spectrum

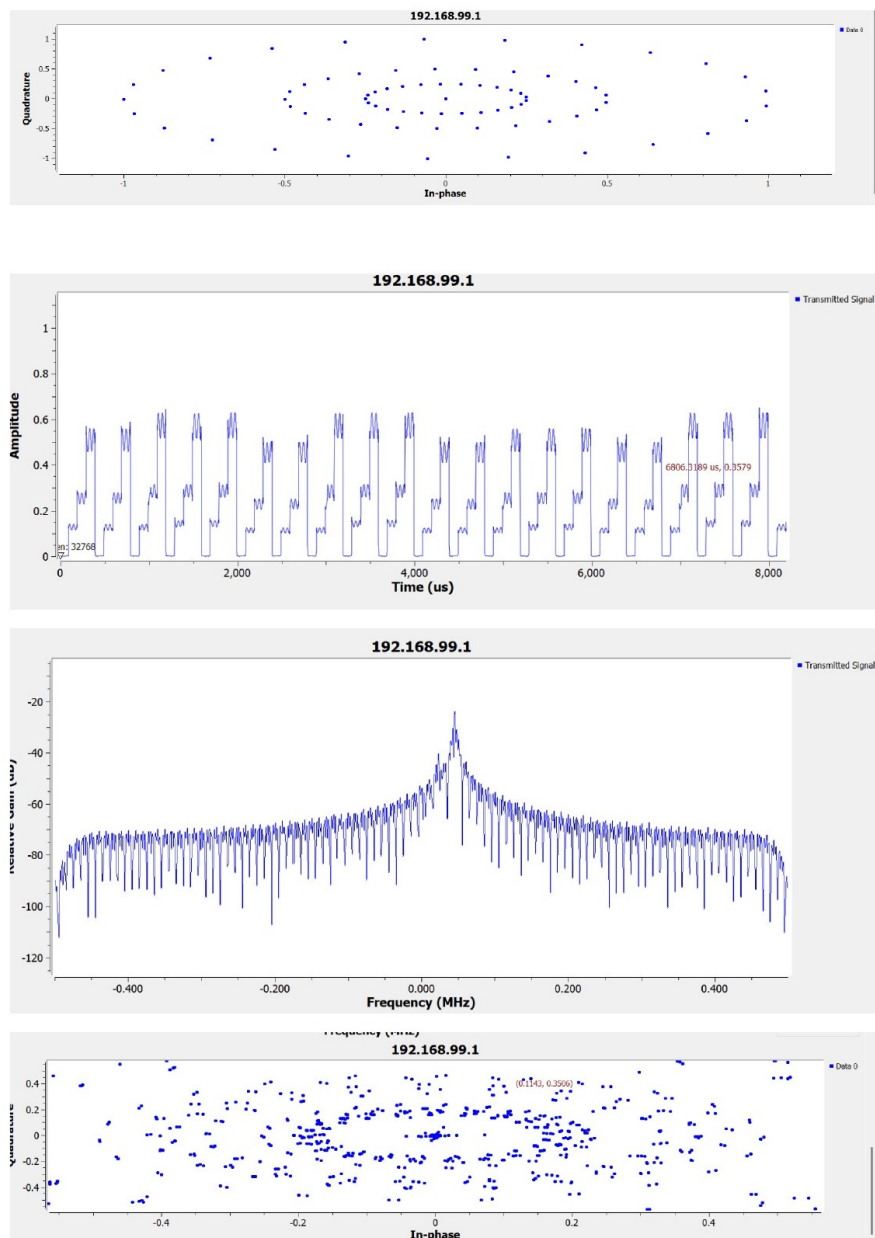
Receiver Side

Fig 6.4(b): ASK receiver — I/Q constellation, demodulated envelope waveform, received power spectrum, and post-AGC constellation scatter

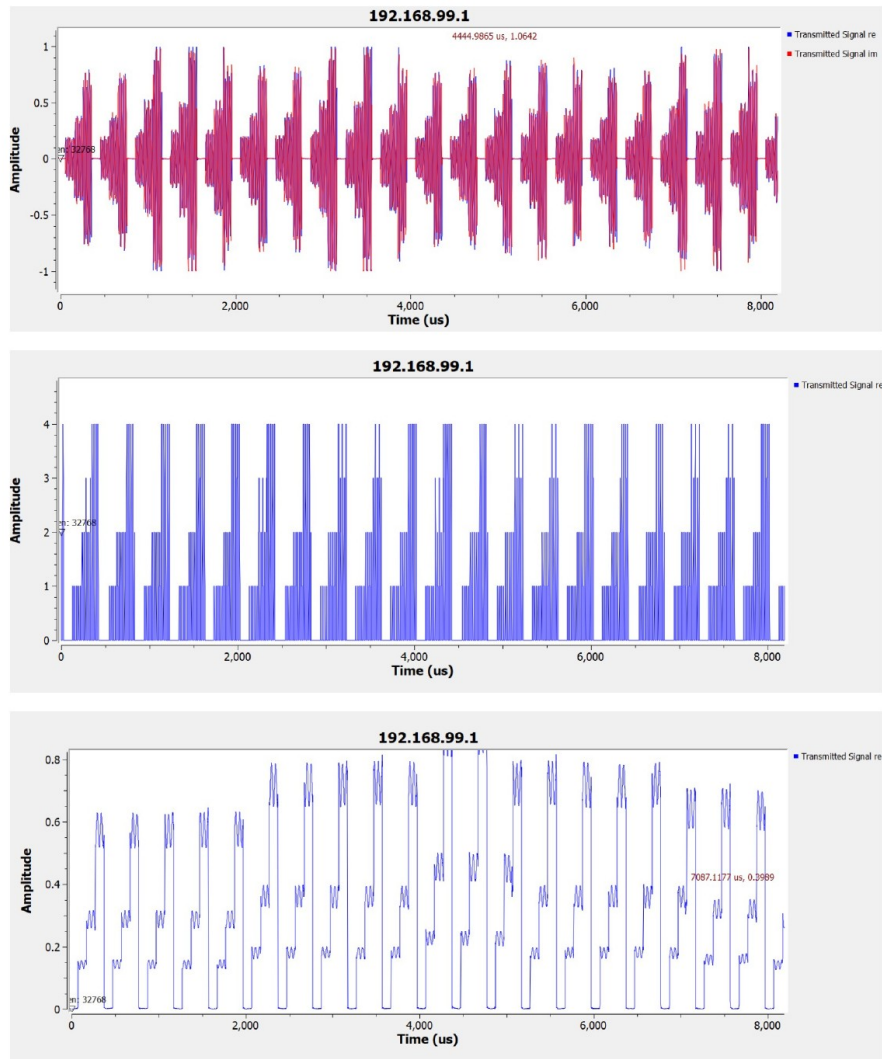


Fig 6.4(c): ASK receiver — additional captures of the demodulated multi-level amplitude waveform

Transmitter Side

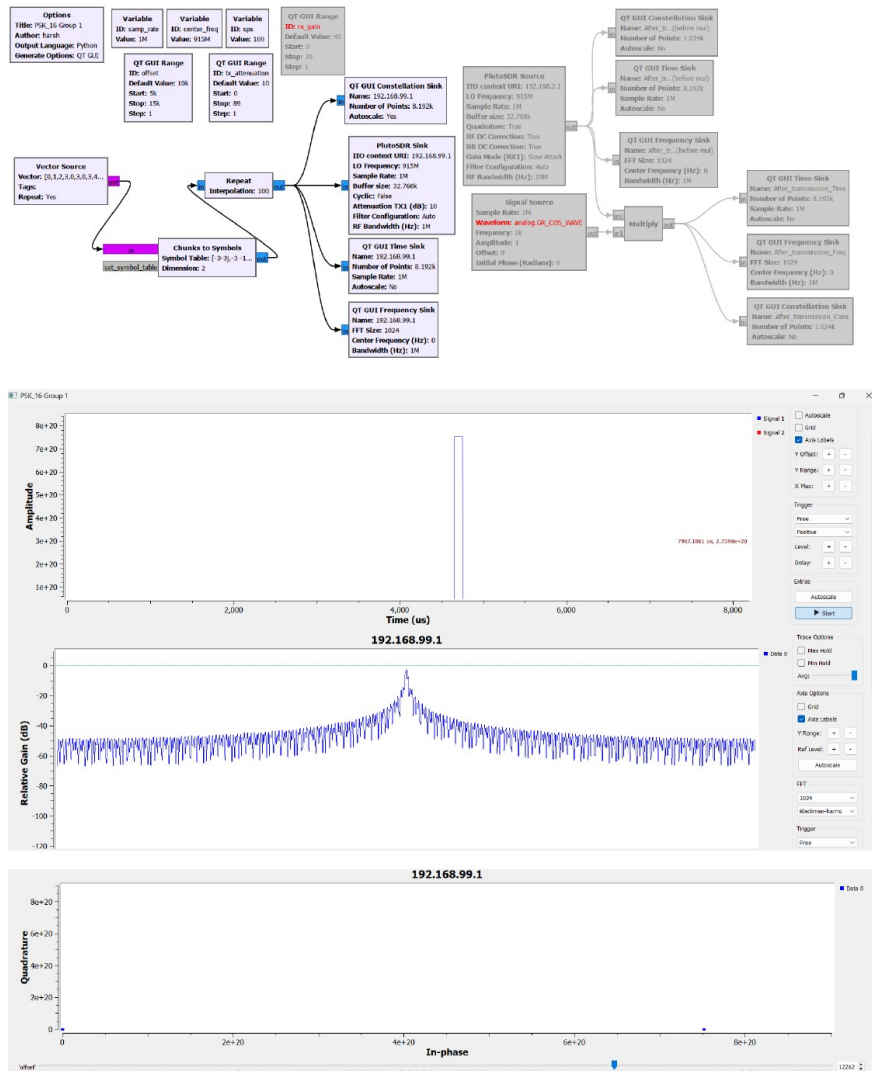


Fig 6.5(a): PSK_16 Group 1 — GRC flow-graph using a 16-point complex Symbol Table, and transmitted baseband waveform / spectrum

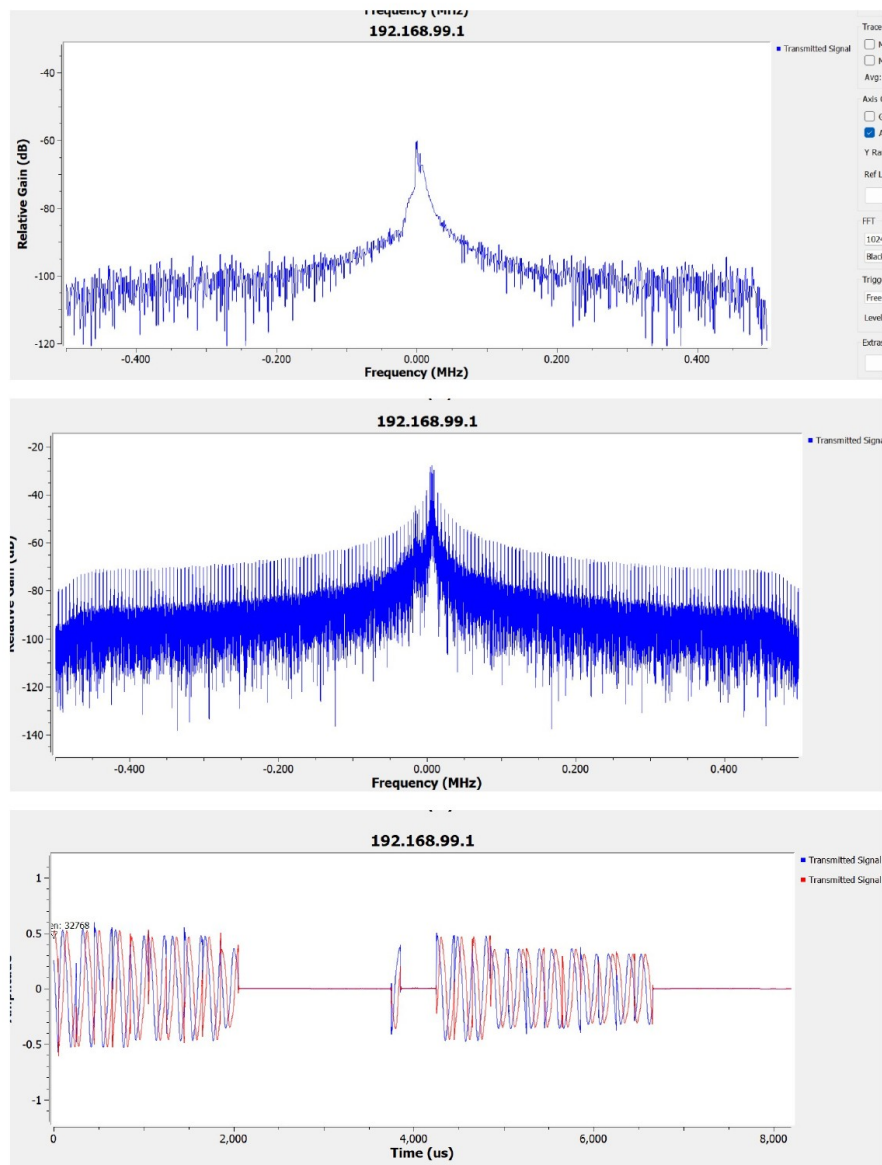
Receiver Side

Fig 6.5(b): 16-level scheme — received power spectrum (two FFT captures) and baseband transmitted waveform

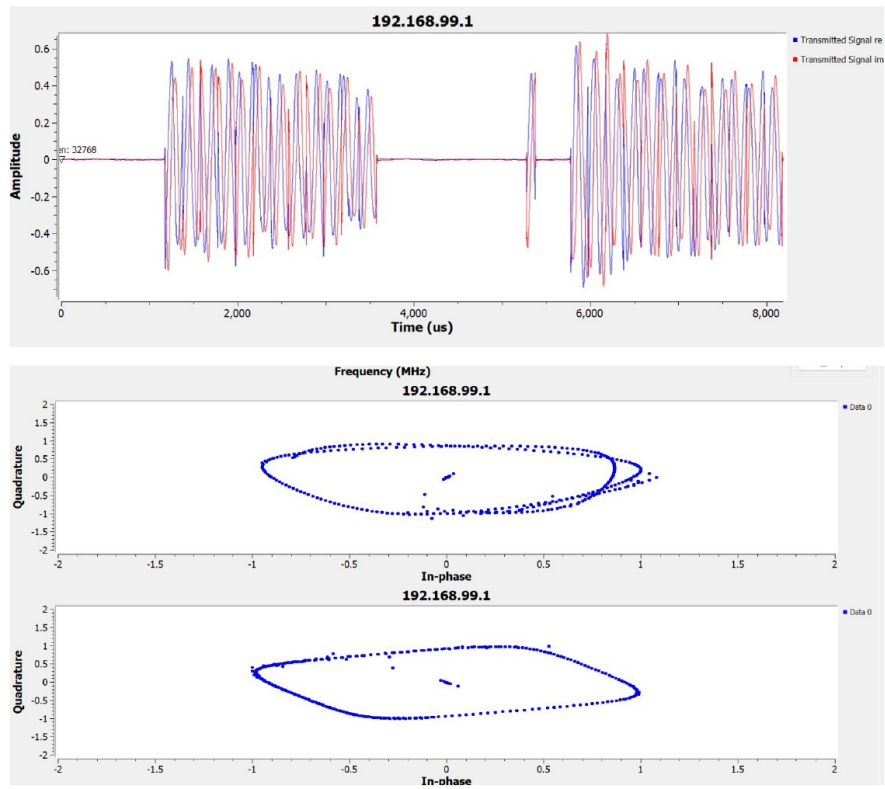


Fig 6.5(c): 16-level scheme — baseband waveform (continued) and final received constellation diagrams (two views)

7. Results & Discussion

7.1 Comparison of Constellation Diagrams

Scheme	Ideal Constellation Points	Observed Pattern
OOK	2 (on real axis: 0 and A)	Recovered as discrete thresholded pulses (1s and 0s) rather than a scatter plot, since detection was done by envelope + threshold, not coherent I/Q demodulation
BPSK	2 (± 1 , 180° apart)	Two/three clustered point groups visible along the real axis; some spread due to channel noise and residual carrier offset
QPSK	4 (90° apart, on a circle)	Points spread into a rotated ellipse instead of 4 tight dots — indicating I/Q gain/phase imbalance in the RF front end
ASK	2-3 discrete amplitude levels on real axis	Scattered points along the real axis with vertical spread, showing amplitude-level overlap caused by noise
16-level scheme	16 points (4x4 grid / 16-QAM style)	Ellipse-shaped ring with points concentrated near the outer edge, similar in shape to the QPSK case, plus an anomalous amplitude overflow observed on the transmitted-signal time plot

7.2 Discussion

The OOK receiver, using simple envelope detection and thresholding, successfully recovered the transmitted bit pattern as clean pulses — this confirms that amplitude-only detection is adequate for a low-order, non-coherent scheme, but such a scheme is fundamentally more vulnerable to noise since a deep fade can be misread as a '0'.

The BPSK constellation showed two/three clusters instead of two mathematically pure points; this scatter is attributed to phase noise, residual frequency offset between the free-running Tx/Rx local oscillators of the two PlutoSDR units, and thermal noise in the wireless channel.

Both the QPSK and the 16-level scheme constellations appeared as an elongated ellipse rather than the expected circular arrangement of points. This is a classic signature of I/Q gain and phase imbalance in the analog front end (unequal gain/phase between the I and Q paths of the direct-conversion PlutoSDR transceiver), which distorts a circle into an ellipse while still preserving the general 'ring' shape of the transmitted symbols.

The ASK constellation showed scattered points along the real (in-phase) axis with a vertical spread that is caused by AWGN and any residual quadrature leakage, but no genuine phase information — consistent with the theory, since ASK conveys information purely in amplitude.

The 16-level (PSK_16) flow-graph exhibited an anomalous, very large amplitude spike (order 10^{20}) on its transmitted-signal time-domain plot. This is not a physically meaningful RF amplitude; it is a numerical/scaling artefact most likely produced inside the Exponentiate/Chunks-to-Symbols processing chain of that particular flow-graph (e.g., an unbounded exponent or an unnormalised constellation causing floating-point overflow) rather than an actual transmitted power level, and should be corrected before drawing further quantitative conclusions from that scheme.

As the number of constellation points/modulation order increases (OOK/ASK $\rightarrow$ BPSK $\rightarrow$ QPSK $\rightarrow$ 16-level), the same channel noise and hardware impairments (I/Q imbalance, phase noise) cause proportionally larger relative errors, because the symbols are packed closer together in the signal space. This matches the general principle that higher-order modulation schemes trade improved spectral efficiency (more bits per symbol) for reduced noise immunity.

7.3 Time and Frequency Domain Observations

In the time domain, OOK and ASK show clear amplitude-level switching in their I/Q waveforms, while BPSK/QPSK show phase-reversal 'kinks' in an otherwise constant-envelope waveform. In the frequency domain, all schemes show a strong carrier-centred lobe with side-lobes caused by the rectangular symbol shaping (no pulse shaping filter was used); higher symbol/interpolation rates and multi-level schemes show a visibly wider or more structured spectral occupancy compared to the simple OOK/BPSK spectra.

8. Conclusion

OOK, ASK, BPSK, QPSK and a 16-level modulation scheme were successfully implemented in GNU Radio Companion and transmitted/received in real time over the air between two ADALM-PLUTO SDR units. Time-domain and frequency-domain analysis confirmed correct modulation behaviour for each scheme, and the constellation diagrams provided a clear visual basis for comparing them: schemes with fewer constellation points (OOK, ASK, BPSK) were more robust and easier to demodulate correctly, while schemes with more points/higher order (QPSK, 16-level) offered higher potential data rate per symbol but showed greater sensitivity to real-world RF impairments such as I/Q imbalance, phase noise and gain mismatch, which appeared as ellipse-shaped or scattered constellations instead of ideal circular/grid patterns. This experiment demonstrates both the practical workflow for building an SDR-based digital communication link using GRC and PlutoSDR, and the fundamental modulation-order versus noise-immunity trade-off central to digital communication theory.